NNN-P-350C
July 16, 1973
SUPERSEDING
Int. Fed. Spec. NNN-P-00350B (GSA-FSS)
August 30, 1967, and
Fed. Spec. NNN-P-350
July 2, 1963

FEDERAL SPECIFICATION

PIPET, MEASURING

This specification was approved by the Commissioner, Federal Supply Service, General Services, Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers requirements for glass measuring pipets suitable for laboratory purposes.

1.2 Classification. Pipets covered by this specification shall be of the following styles and capacities as specified (See 6.1).

1.2.1 Styles and Capacities.

Styles.

Style I - Standard Taper Tip
Style II- Long Taper Tip

Capacities.

0.1 milliliter (ml.) (0.01 ml. subdivisions)
0.1 ml. (0.001 ml. subdivisions, Style II only)
0.2 ml. (0.01 ml. subdivisions)
0.2 ml. (0.001 ml. subdivisions, Style II only)
0.5 ml. (0.01 ml. subdivisions, Style II only)
1 ml. (0.1 ml. subdivisions, Style II only)
1 ml. (0.01 ml. subdivisions)
2 ml. (0.1 ml. subdivisions)
2 ml. (0.01 ml. subdivisions, Style II only)
5 ml. (0.1 ml. subdivisions)
10 ml. (0.1 ml. subdivisions)
25 ml. (0.1 ml. subdivisions, Style I only)

2. APPLICABLE DOCUMENTS

2.1. The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Boxes, Shipping, Fiberboard.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

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(Single copies of this specification and other Federal Specifications required by ctivities outside the Federal Government for bidding purposes are available without harge from Business Service Centers at the General Services Administration Regional ffices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, ort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, tandards, and Handbooks and the Index of Federal Specifications and Standards from stablished distribution points in their agencies.)

Military Specification:

MIL-L-10547 - Liners, Case, and Sheet, Overwrap, Water-Vaporproof, or Waterproof, Flexible.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by suppliers in connection ith specific procurement functions should be obtained from the procuring activity r as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification o the extent specified herein. Unless a specific issue is identified, the issue n effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (NBS) Publication:

NBS Circular 602 - Testing of Glass Volumetric Apparatus.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402).

American Society for Testing and Materials (ASTM) Standards:

C 148 - Polariscopic Examination of Glass Containers.
C 162 - Definition of Terms Relating to Glass and Glass Containers.
E 438 - Glasses in Laboratory Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section. 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Material. Pipets shall be made of glass conforming to the requirements of type I, classes A or B (Borosilicate Glass), or type II (Soda Lime Glass) of ASTM E 438 as specified (see 6.1).

3.2 Annealing. When the pipets are examined with a polariscope as specified in 4.4.1 there shall be no ring strain greater than temper No. I except as permitted for the tempered tip in 3.6. Longitudinal strain, if present, shall be faint and slightly diffused.

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3.3 Design.

3.3.1 Shape. Pipets shall be straight and of one piece. Any cross section of the pipet taken in a plane perpendicular to the longitudinal axis shall be circular. The shape shall permit complete emptying and thorough cleaning. The 5-, 10- and 25-milliliters (ml) pipets shall have constricted mouthpieces that are reduced in diameter, length 25 ± 5 millimeter (mm); outside diameter 7.5 ± 1.5 mm. for 5- and 10-ml. pipets, and 8.5 ± 1.5 mm. for the 25 ml. pipets. The outside edge of the top shall be beveled, and the bevel shall be ground or fire-polished. The top surface shall be ground or fire-polished.

3.3.2 Delivery tips. Delivery tips of the style I pipets shall be made with a gradual taper of 20-30 mm. for capacities up to 5 ml. inclusive, and 25-40 mm. for the 10 and 25 ml. capacity pipets. Style II pipets shall have a taper at a distance of 60-65 mm. from the tip end. The outside diameter (OD) of the tip shall be 2.5-3.5 mm. for a distance of 10 mm. measured from the tip end. The tip end shall be perpendicular to the longitudinal axis of the pipet. The tip shall be ground or fire polished.

3.3.3 Dimensions. The pipets shall conform to the essential dimensions given in Table I.

TABLE I. Requirements and Tolerances for Measuring Pipets

Capacity (ml)		Graduations (ml)		Scale (mm)		Outflow time (seconds)		Color Band
Capacity	accuracy tolerance	Least value	Main numbered	Length	Outside diameter minimim (min.)	Min.	Max.	
0.1	±0.005	0.01	0.01	100 to 150	5.5	0.5	10	white
		0.001	0.01	100 to 150	5.5	0.5	10	2-green
0.2	±0.008	0.01	0.02	110 to 160	5.5	0.5	8	black
		0.001	0.02	175 to 250	5.5	0.5	8	2-blue
0.5	±0.01	0.01	0.1	140 to 200	6.0	2.	6	2-yellow
1.	±0.02	0.1	0.1	135 to 210	6.0	3.	22	red
		0.01	0.1	135 to 210	6.0	3.	22	yellow
2.	±0.02	0.1	0.2	160 to 210	7.0	3.	22	green
		0.01	0.2	160 to 210	7.0	3.	22	2-white
5.	±0.04	0.1	1.0	160 to 210	7.5	8.	20	blue
10.	±0.06	0.1	1.0	180 to 240	9.5	8.	30	orange
25.	+0.10	0.1	1.0	250 to 300	12.0	15.	35	white

3.4 Markings. All markings shall be permanent and legible. Markings shall be applied by one of the following methods - etched and filled with a permanent pigment; or by application of a stain which is fired, into the glass without etching; or by application of an enamel which is fused onto the glass without etching. Pigment or stain shall be dark in color. Marking shall not change in appearance when tested for permanency of marking as specified in 4.4.3.

3.4.1 Graduation markings. Main graduation markings shall extend a minimum of 3/4 of the way around the pipet; intermediate graduations shall extend a minimum of 1/5 of the way around the pipet; and the least graduation markings shall extend a minumum of 1/7 of the way around the pipet. The value of all main graduations shall be inscribed in arabic numerals directly above the line referenced. Graduation markings shall be in a plane perpendicular to the longitudinal axis of the pipet and parallel to each other. The graduation mark shall be a minimum of 80 mm. from the top of 0.5 ml. pipet, and a minimum of 90 mm. from the top of all other pipets.

3.4.2 For identification. Each pipet shall be marked with the manufacturer's name or trademark, its capacity, the symbol "TD", and the temperature of calibration (20°C.)

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3.5 Color coding. Each pipet shall be marked (see table I), with one or two bands as specified. For a code consisting of a single band, the band shall be six to ten mm. wide and for a code consisting of a double band, each band shall be three to five mm. wide and separated by a space of two to three mm. The colored area shall cover at least 40 percent of the circumference of the pipet and does not have to be continuous but may be broken. It shall be situated not more than 40 mm. from the top of the pipet. The color of the band shall be bright and distinct. The color shall show no damage other than loss of gloss when tested as specified in 4.4.3. Banding shall meet the alkali and acid tests as specified in 4.4.3.1 and 4.4.3.2, respectively.

3.6 Tempered tips. Each pipet shall have a tempered tip. When tested as specified in 4.4.4., the optical retardation for Style I tips shall be in the range of 100 to 200 millimicrons (mμ), and for Style II tips in the range of 50 to 200 mμ.

3.7 Outflow. The rate of outflow of the pipet shall be as shown in table I when tested as specified in 4.4.5.

3.7.1 Workmanship. The pipet shall be free from defects which impair serviceability. The pipet shall be transparent, clean, dry, and free of foreign material. The bore of the pipet shall be free of constrictions and shall permit complete emptying of fluids.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. As used in this specification, inspection is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties of the item).

4.1.2 Certificates of analysis. In lieu of contractor performing tests, certified test data furnished by the manufacturer of the glass will be acceptable as evidence of conformance to 3.1.

4.1.3 Size of lot. A lot shall consist of all pipets of the same classification which have been manufactured by the one manufacturer and offered for acceptance at one time. The pipets shall be manufactured under essentially the same process, and at essentially the same time.

4.1.4 Records. Records of examination and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of two years after delivery of the supplies to which such records relate.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II.

TABLE II. Sampling for examination

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-3	4.0

4.2.2 For tests. Sampling for tests shall be in accordance with MIL-STD-105. Unit of products shall be pipet.

4.2.3 Test solution. Where possible, the same sample unit shall be used for the determination of two or more test characteristics in accordance with table III.

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TABLE III. Testing requirements

Characteristic	Requirement paragraph	Test procedure	Inspection level	AQL (percent) defective)
Annealing	3.2	4.4.1	S-2	1.0
Capacity	3.4	4.4.2	S-2	1.0
Permanency of marking		4.4.3		
Alkali test	3.5	4.4.3.1	S-1	4.0
Acid test	3.5	4.4.3.2	S-1	4.0
Tempered tips	3.6	4.4.4	S-2	1.0
Outflow	3.7	4.4.5	S-2	1.0

4.3 Examination. The pipets shall be examined to determine compliance with all the requirements contained in the specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with Table IV.

TABLE IV. Classification for defects

Categories	Defect 1/
Major	
101	Pipet not circular, free of distortion, or obstructions.
102	Pipet not free of burrs, protrusions, or sharp edges.
103	Pipet not free of cracks or chippings.
104	Pipet not free of stones or blisters.
105	Pipet glass not clear.
106	Pipet not free of grease or fired in foreign matter.
107	Capacity mark on pipet not present, not perpendicular to longitudinal axis, or not distinct.
108	Color band on pipet incorrect, or not present.
109	Mouthpiece and top of pipet not beveled, ground or fire-polished.
110	Marking on pipet not complete, correct, distinct or legible including manufacturer's identification.
Minor	
201	Pipet glass not free of scratch marks.
202	Pipet not free of bubbles.

1/ Examination shall not be restricted to the classified defects listed above. Definition of glass defects shall be in accordance with ASTM C162.

4.3.2 Dimensional examination. The pipets shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Distilled water and reagent grade chemicals shall be used throughout the test. Test shall be conducted as follows:

4.4.1 Annealing. Pipets shall be tested in accordance with method A or B of ASTM C148.

4.4.2 Capacity. Pipets shall be tested for capacity accuracy in accordance with the test procedures of Section 3.5 of NBS Circular 602 or equivalent.

4.4.3. Permanency of marking. A different pipet shall be used for each of the following tests.

4.4.3.1 Resistance to alkali. The pipet shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 1/2 hour, and allowed to stand in the alkali for an additional hour. The pipet shall be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

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4.4.3.2 Resistance to acid. The pipet shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled, for 1/2 hour, and allowed to stand in the acid for an additional hour. The pipet shall then be removed, thoroughly rinsed with distilled water, and dried by rubbing vigorously with a cloth.

4.4.4 Tempered tips. Tips shall be tested in accordance with method A or B of ASTM C148. The tips shall be immersed in a solution whose index if refraction matches that of the glass.

4.4.5 Outflow time. Outflow time of the pipets shall be determined by means of a stopwatch to the standards as shown in table I.

4.5 Examination of preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements in section 5. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery with the exception that it need not be sealed. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE V. Preparation for delivery defects

Examine	Defects
Materials	Any component missing, damaged, or not specified.
Packaging	Unit packaging not as specified.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping (when required), case liner not properly closed (when required), container bulged or distorted.
Contents	Quantity in interior packages and shipping containers not as specified.
Markings	Omitted, incorrect, illegible, of improper size, location, sequence, or method of application.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A, B, or C as specified (See 6.1).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each pipet shall be packaged in a spirally wound fiber or paperboard tube or in a carton of suitable size and design. The tube or carton shall be adequately secured to prevent accidental opening, shifting, or damage to, the contents.

5.1.1.2 Intermediate packaging. Eighteen pipets (See 6.1) of all sizes up to 25 ml. of Style I and of the same capacity; twelve pipets of Style I, 25 ml.; and twelve pipets of Style II and the same capacity shall be packed in a box conforming to PPP-B-636, type CF, class domestic, grade 200. The box shall be designed with suitable interior fittings, or shall contain sufficient cushioning material to insure against shifting, or of damage to the contents.

5.1.2 Level B.

5.1.2.1 Unit packaging. Six pipets of the same style and capacity shall be packed in the manner normally used for the item.

5.1.2.2 Intermediate packaging. All pipets up to 25 ml. of Style I of the same size capacity shall be intermediate packaged 18 pieces (pcs). in a shelf pack, the 25 ml. of Style I pipet shall be intermediate packaged 12 pcs to a shelf pack, and Style II pipets of the same size and capacity shall be intermediate packaged 12 pcs to a shelf pack as specified in 5.1.1.2.

5.1.3 Level C. Pipets shall be packaged in accordance with the manufacturer's commercial practice suitable for shipment from the supply source to the first receiving activity for immediate use.

5.2 Packing. Packing shall be level A, B, or C, as specified (See 6.1).

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5.2.1 Level A. Sixteen intermediate packages shall be packed in a box conforming to PPP-B-621, class 1, Table II, PPP-B-601, overseas type, or PPP-B-636, type CF, class weather resistant, grade 275. Each wooden box shall be fitted with a water-proof case liner conforming to MIL-L-10547. As far as practical, exterior containers shall be of uniform size and shape, shall be of minimum cube and tare, and contain identical quantities of like items. Gross weight shall not exceed the weight limitations of the applicable box specification of appendix thereto. Strapping, when required, shall be in accordance with the applicable box specification. (Note: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.)

5.2.2 Level B. Sixteen intermediate packages of the same style and capacity shall be packed in a fiberboard box conforming to PPP-B-636, type CF, class domestic, grade 275.

5.2.3 Level C. Pipets of the same style and capacity in quantities as specified (See 6.1), packaged as specified in 5.1 shall be packed to insure carrier acceptance and safe delivery at first destination. Containers shall comply with the Uniform Freight Classification Rules, or National Motor Freight Classification Rules, as applicable to the mode of transporation.

5.3 Marking. In addition to any special markings required by the contract order, marking of the interior packages and exterior shipping containers shall be in accordance with Fed. Std. No. 123 for civil agencies and MIL-STD-129 for military agencies, as applicable (See 6.1). All unit packages, intermediate packages, and shipping containers shall be marked to show the name of the item, the Federal Stock Number, the quantity contained therein, and "Glass-Handle with Care."

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Material required (See 3.1).
(c) Style, and capacity required (See 1.2).
(d) Level of packaging and packing required (See 5.1 and 5.2).
(e) Quantities required (See 5.2.3).
(f) Marking required (See 5.3).

MILITARY CUSTODIANS:

Army - MU
Navy - MS
Air Force - 03
DSA - DM

Review activity:

DSA - DM

Preparing activity:

GSA-FSS

Civil Agency Coordinating Activities:

GSA-FSS
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